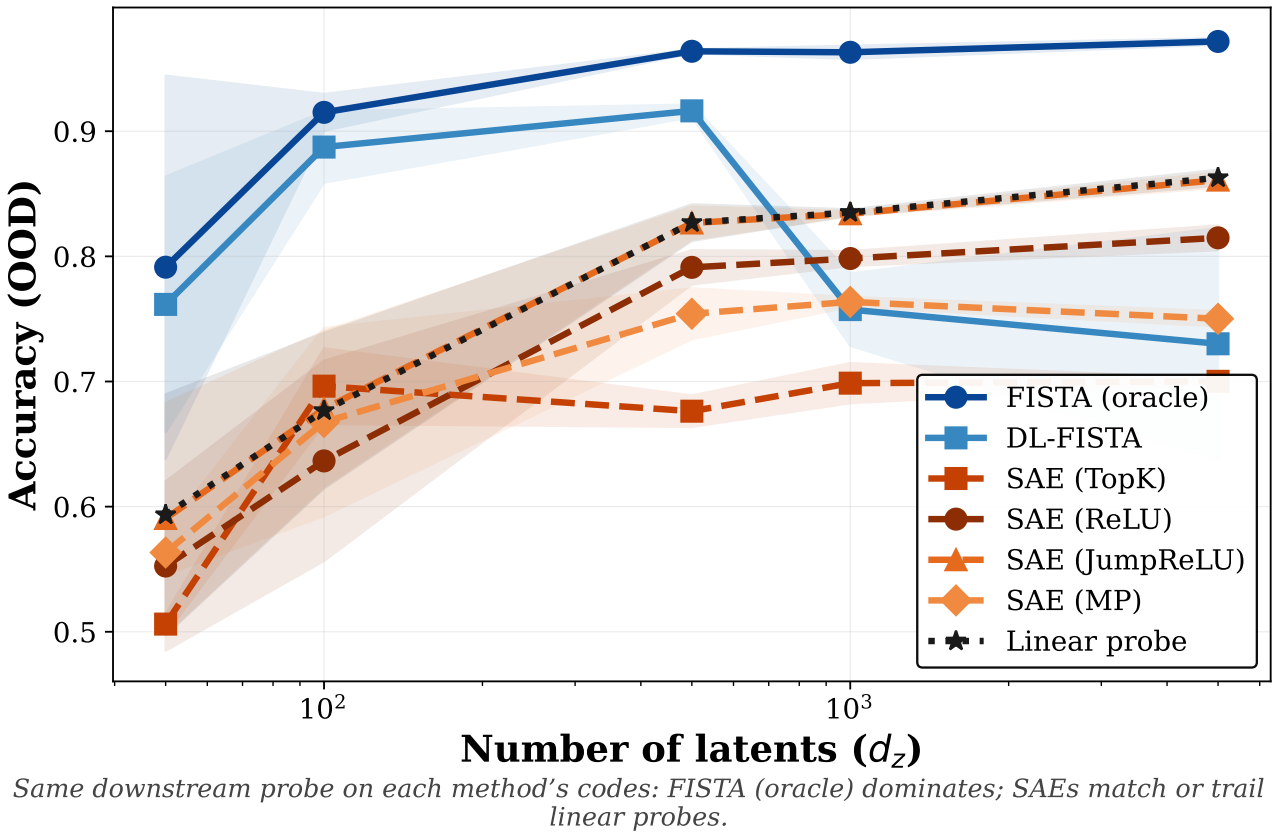

Figure 24: **Varying latent dimension: Accuracy (OOD).** Duplicate of Figure 7 for completeness alongside other appendix metrics.

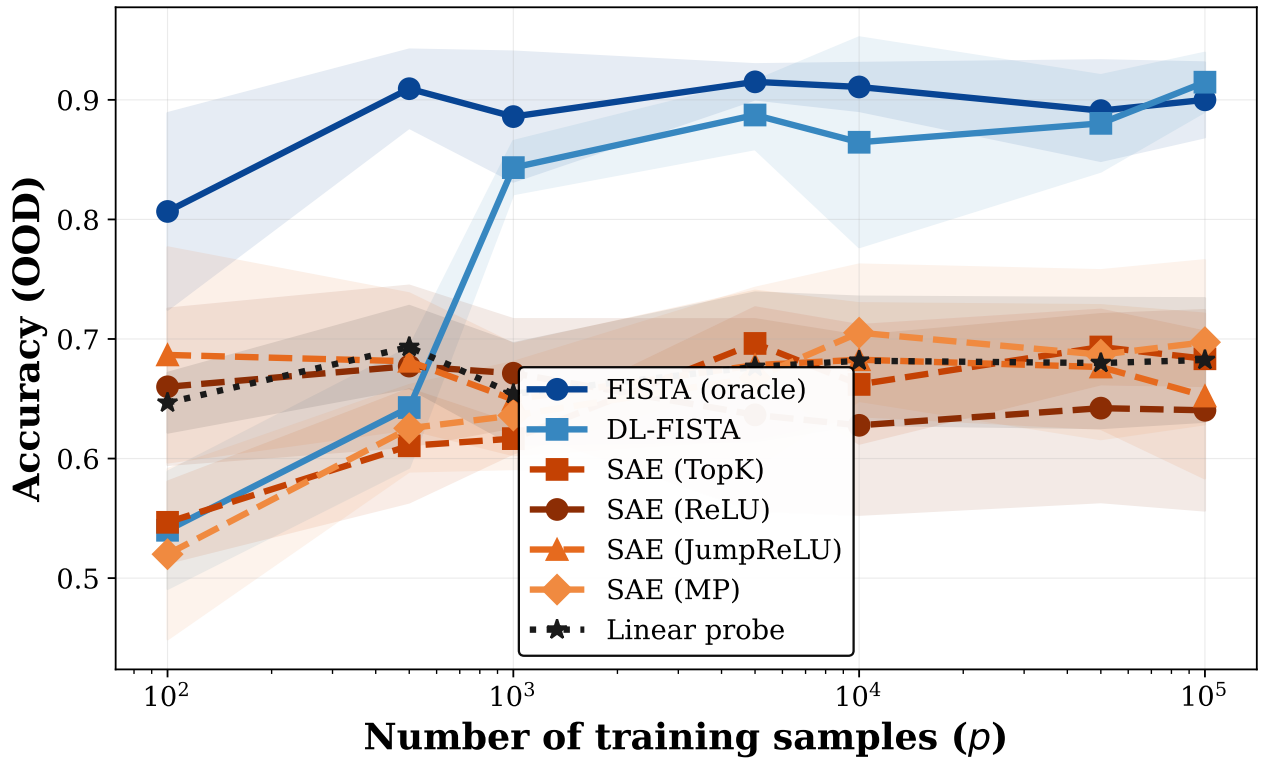

Figure 25: **Varying training data: Accuracy (OOD).** Duplicate of Figure 11 for completeness alongside other appendix metrics.

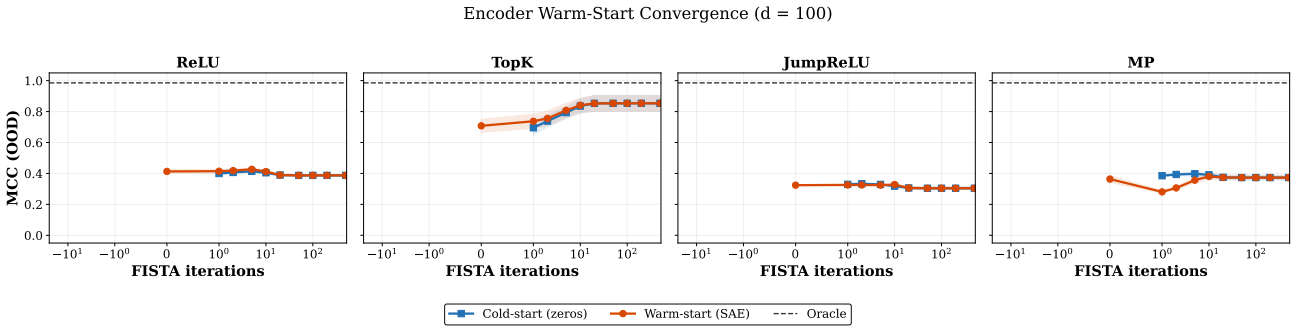

Figure 26: **Encoder warm-start convergence** ($d_z = 100$)**.** Cold-start (blue) and warm-start (orange) FISTA on frozen SAE decoder. The convex Lasso objective means both converge to the same optimum. Warm-starting provides a modest advantage for TopK at low iteration budgets but negligible benefit for other types.

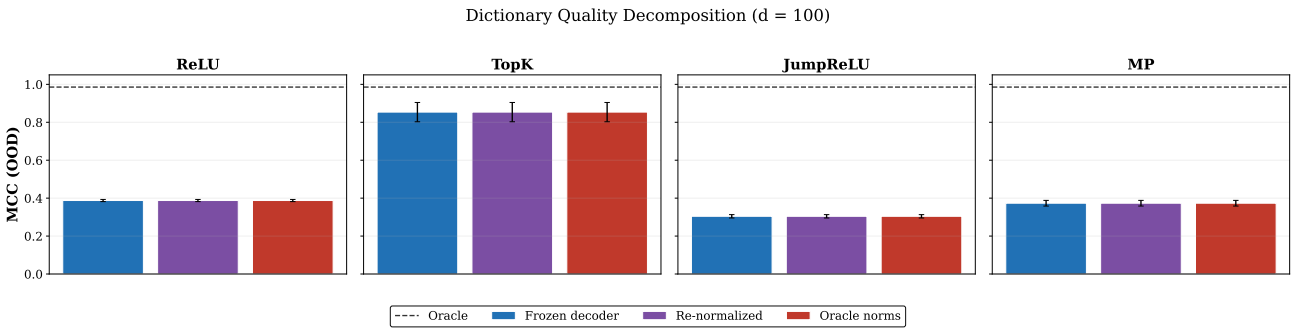

Figure 27: **Dictionary quality decomposition** ($d_z = 100$)**.** Blue: FISTA with frozen SAE decoder. Purple: after re-normalising decoder columns to unit norm. Red: SAE directions with oracle column magnitudes. Re-normalising and norm substitution have no effect—the error is in the column *directions*, not magnitudes.

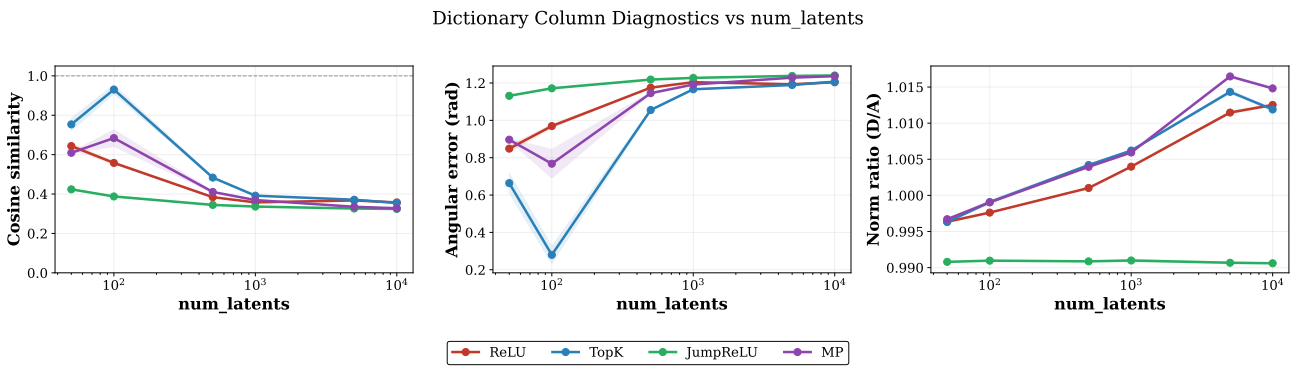

Figure 28: **Dictionary column diagnostics across** d_z**.** Left: mean cosine similarity between SAE and ground-truth columns. Centre: angular error. Right: norm ratio. TopK maintains high cosine (> 0.9) across all d_z; other types degrade. Norm ratios are $\approx$1.0 throughout, confirming the error is directional.

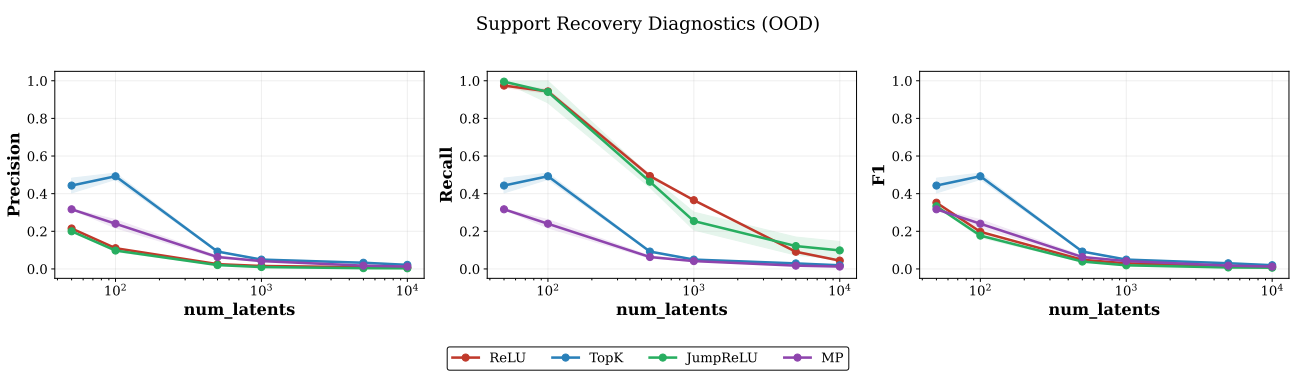

Figure 29: **Support recovery diagnostics across** d_z**.** Precision, recall, and F1 of the SAE's binary support compared to ground truth. ReLU and JumpReLU have high recall but catastrophically low precision (~0.1): they activate nearly all features. TopK maintains balanced precision and recall (~0.5).